

A Digit-Concatenation Puzzle

Complete Solution

Problem 1. Let A be a two-digit number and B a three-digit number such that the product $C = A \cdot B$ is a five-digit number and the ten digits of A, B, C taken together are exactly $0, 1, \dots, 9$ (each used once). Suppose furthermore that

$$A < R(A), \quad B > R(B), \quad B + R(B) = 11(A + R(A)),$$

where $R(x)$ denotes the integer obtained by reversing the digits of x . Determine the ten-digit number N formed by concatenating A, B , and C .

Notation and First Reductions

Write

$$A = 10a + b, \quad B = 100c + 10d + e,$$

where a, b, c, d, e are pairwise distinct digits and $a, c \neq 0$. Reversing gives

$$R(A) = 10b + a, \quad R(B) = 100e + 10d + c.$$

The inequalities reduce to

$$A < R(A) \iff 10a + b < 10b + a \iff a < b, \quad B > R(B) \iff 100c + 10d + e > 100e + 10d + c \iff c > e.$$

Put $s = a + b$. Then

$$A + R(A) = 11(a + b) = 11s,$$

and the given reversal relation $B + R(B) = 11(A + R(A))$ becomes

$$B + R(B) = 101(c + e) + 20d = 121s. \tag{1}$$

Two congruences from (1)

Reducing (1) modulo 10 gives

$$c + e \equiv s \pmod{10}. \tag{2}$$

Reducing (1) modulo 11, using $101 \equiv 2, 20 \equiv 9, 121 \equiv 0 \pmod{11}$,

$$2(c + e) + 9d \equiv 0 \pmod{11} \iff c + e \equiv d \pmod{11}. \tag{3}$$

Lemma 1. $a + b = d = c + e$.

Proof. Since $0 \leq c + e \leq 17$ and $0 \leq d \leq 9$, congruence (3) leaves only

$$c + e = d \quad \text{or} \quad c + e = d + 11.$$

If $c + e = d + 11$, substituting into (1) gives

$$101(d + 11) + 20d = 121s \iff 121d + 1111 = 121s \iff 1111 = 121(s - d),$$

impossible, because $1111 \equiv 22 \pmod{121}$ is not divisible by 121. Hence $c + e = d$, and (1) becomes $121d = 121s$, so $s = d$. $\square$

Consequently all six digits a, b, c, d, e and 0 are forced to be distinct where they occur: indeed $e \neq 0$ (otherwise $c + e = d$ forces $c = d$, contradicting distinctness), $b \neq 0$ since $a < b$, and $a, c \neq 0$ by hypothesis. Therefore the digit 0 must appear in C .

Modulo 9: Narrowing the Middle Digit

The digit sums are

$$A : a + b = d, \quad B : c + d + e = (c + e) + d = 2d, \quad C : 45 - 3d,$$

because all ten digits occur exactly once. Since $C = A \cdot B$ and every number is congruent to the sum of its digits modulo 9,

$$45 - 3d \equiv d \cdot 2d = 2d^2 \pmod{9} \iff d(2d + 3) \equiv 0 \pmod{9}.$$

As $d = c + e$ with $c > e \geq 1$ and $d \leq 9$, we have $3 \leq d \leq 9$. Checking $d = 3, \dots, 9$:

$$d = 3 : 3 \cdot 9 = 27 \equiv 0, \quad d = 6 : 6 \cdot 15 = 90 \equiv 0, \quad d = 9 : 9 \cdot 21 = 189 \equiv 0,$$

while $d = 4, 5, 7, 8$ give residues 8, 2, 2, 8. Hence

$$d \in \{3, 6, 9\}.$$

Case Analysis

Case $d = 3$.

$c + e = 3$ with $c > e \geq 1$ forces $(c, e) = (2, 1)$, so $B = 231$. Then $a + b = 3$ with $a < b$ forces $A = 12$, but the digits 1, 2 already occur in B . No solution.

Case $d = 6$.

$c + e = 6$ with $c > e \geq 1$: $(c, e) = (5, 1)$ or $(4, 2)$.

- $B = 561$: $a + b = 6$ with $a < b$ and digits distinct from $\{5, 6, 1\}$ leaves only $A = 24$ (the pairs $(0, 6)$ and $(3, 3)$ are excluded by $a \neq 0$ and distinctness). But $24 \cdot 561 = 13464$ has repeated digits. Rejected.
- $B = 462$: $a + b = 6$ leaves $A = 15$, but $15 \cdot 462 = 6930$ is not a five-digit number. Rejected.

Case $d = 9$.

$c + e = 9$ with $c > e \geq 1$: $(c, e) \in \{(8, 1), (7, 2), (6, 3), (5, 4)\}$, so

$$B \in \{891, 792, 693, 594\},$$

and $a + b = 9$ with $a < b$ gives

$$A \in \{18, 27, 36, 45\}.$$

Discard candidates by two tests: (i) any digit of A already occurring in B ; (ii) the units digit of C , which is $be \bmod 10$, colliding with a used digit.

B	A after test (i)	outcome
891	27, 36, 45	units digit of C equals $b \in \{7, 6, 5\}$, already used — all fail
792	18, 36, 45	$18 \cdot 792 = 14256$ repeats 1, 2; 36: units $6 \cdot 2 \equiv 2 \equiv 2$ used; $45 \cdot 792 = 35640$ repeats 4, 5 — all fail
693	18, 27, 45	$18 \cdot 693 = 12474$ has repeated digits; $27 \cdot 693 = 18711$ repeats; 45: units $5 \cdot 3 \equiv 5$ used — all fail
594	18, 27, 36	$18 \cdot 594 = 10692$ repeats 1, 9; 36: units $6 \cdot 4 \equiv 4$ used; $27 \cdot 594 = 16038$ works

Verification and Answer

$$27 \cdot 594 = 16038, \quad 27 < 72, \quad 594 > 495,$$

$$594 + 495 = 1089 = 11 \cdot 99 = 11(27 + 72).$$

All conditions hold, and

$$N = 27 \mid 594 \mid 16038 = 2759416038$$

uses every digit 0–9 exactly once.

Answer 1.

$$A = 27, \quad B = 594, \quad C = 16038, \quad N = 2759416038.$$